

Nonlinear Distortion of Semiconductor Optical Amplifiers as a Source of Equalization Enhanced Phase Noise

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Abstract We demonstrate the interaction between nonlinear phase distortion induced by semiconductor optical amplifiers (SOA) and digital equalization via both simulations and experiments. Properties of the resulting equalization enhanced phase noise (EEPN) are also compared with the well-studied EEPN caused by laser phase noise. ©2026 The Author(s)

Introduction

Semiconductor optical amplifiers (SOA) present a promising solution for signal amplification in optical communication systems owing to its compactness, cost-effectiveness, and design flexibility. However, their deployment is significantly constrained by the inherent nonlinear distortion. In particular, the extra phase noise (PN) induced by SOAs may result in significant performance penalty in coherent systems especially when high-order quadrature amplitude modulation (QAM) formats are employed. Abundant research has analyzed the SOA-induced nonlinear distortion and the mitigation techniques [1-6]. However, unlike the concept of equalization enhanced phase noise (EEPN) related to laser phase noise (LPN) [7], the topic of how digital signal processing (DSP) interplays with the SOA-induced PN has not been fully explored.

In this work, we provide both numerical and experimental demonstration of the SOA-PN-induced EEPN whose manifestation is determined by the location of SOAs and digital equalization. We show that while the initial SOA nonlinearity mainly manifests as phase distortion, its interaction with DSP potentially generates more amplitude noise and further degrades system performance. Moreover, we summarize several key properties of the SOA-PN-induced EEPN that are different from the well-understood LPN-induced EEPN.

Principle and Simulation

In this work we assume a dual-polarization (DP) coherent system with 16-QAM format, and an SOA is employed for each polarization of signal. To simulate the SOA behavior, we follow the equations widely adopted in previous publications, which link the input and output signal field as

$$E_{out}(t) = E_{in}(t) \exp\left[\frac{h(t)}{2}(1 + j\alpha)\right] \quad (1)$$

where α is the SOA linewidth enhancement factor, and $h(t)$ is the material gain exponent which is determined by

$$\tau_c \frac{d}{dt} h(t) = h_0 - h(t) - [e^{h(t)} - 1] \frac{|E_{in}(t)|^2}{P_{sat}} \quad (2)$$

where τ_c , h_0 and P_{sat} are the carrier life time, unsaturated material gain exponent and saturation power of the SOA, respectively. Furthermore, we use a sufficient over-sampling rate combined with the Runge-Kutta method when simulating SOA to improve the model accuracy. The SOA effects not included in Eq. (1) and Eq. (2) are not simulated.

As indicated in Eq. (1) and Eq. (2), nonlinear distortion is introduced as the instantaneous SOA gain depends on the input signal pattern, and it contains more phase error with typical values of α . Therefore, to better reveal the noise characteristics, we evaluate the strength of distortion in amplitude and phase dimensions separately using specific symbols, as follows (also visually summarized in Fig. 1): (1) the 16-QAM symbols after receiver DSP are normalized such that the constellation points in either real or imaginary dimension have mean values of $\{-3, -1, 1, 3\}$; (2) the symbols corresponding to the four corner 16-QAM constellation points are selected; (3) the differences between the radii of these symbols and a reference radius (i.e., $3\sqrt{2}$) are calculated, and their root-mean-square (RMS) value is computed (corresponding to the “amplitude noise” in the following analysis); (4) similarly, the RMS value of errors in the arc length dimension is calculated for these symbols (corresponding to the “phase noise” in the noise analysis). Next, we consider two specific configurations where EEPN may be induced.

Case 1: Chromatic dispersion (CD) is pre-compensated in transmitter (Tx) DSP, and SOAs are also integrated within the transmitter. In this case, the SOA-induced PN undergoes a non-zero net CD, which, in principle, results in EEPN. Note

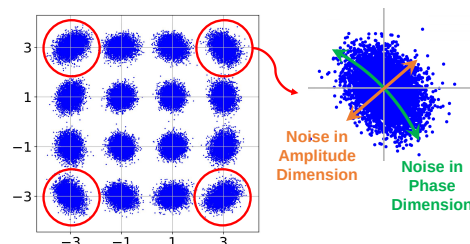


Fig. 1: Illustration of the employed noise analysis method.

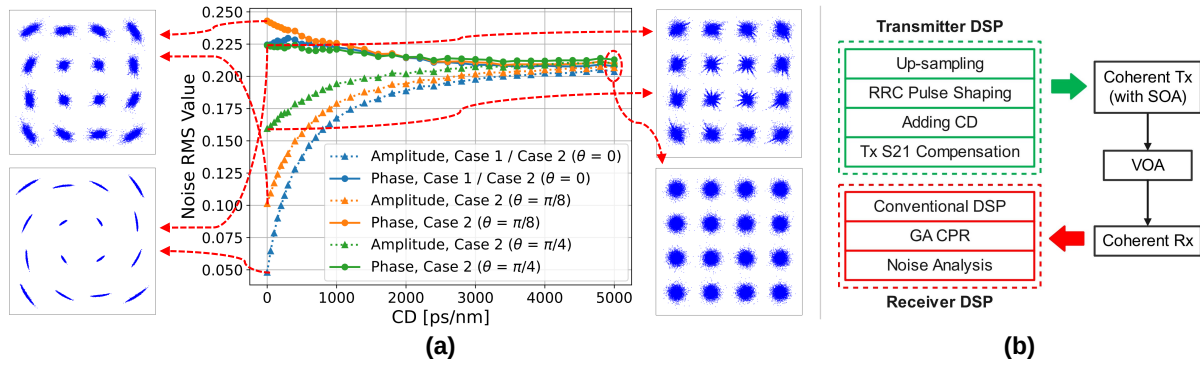


Fig. 2: (a) Noise analysis results in different simulation configurations, and (b) experimental setup.

that in this configuration, each SOA generally aligns with one signal polarization. As a result, SOA-induced distortion in the symbols after receiver DSP is basically determined by the transmitted signal pattern of the same polarization.

Case 2: Both CD compensation and SOAs are placed at receiver (Rx), so the SOA-induced PN also experiences a non-zero net CD and leads to EEPN. However, there may be a more complicated interaction in this case between signal, SOA non-linearity and DSP after considering polarization-related effects added by the channel such as polarization rotation. For example, the transmitted signal of one polarization may practically get split into two orthogonal polarizations at the Rx SOA input and collect different pattern-dependent PN, which causes imperfect re-construction of the original signal by DSP. This is different from the LPN-induced EEPN, where LPN acts as a common impairment for both polarizations, and the corresponding EEPN is generally independent of polarization rotation.

To verify the existence of EEPN, simulations were first conducted at a symbol rate of 87.5 Gbaud considering SOA nonlinearity, CD, and polarization rotation as potential impairments. The SOA of each polarization had an output power of 3.3 dBm, with other parameters set as: $\alpha = 5$, $\tau_c = 200$ ps, $h_0 = 4.14$, and $P_{sat} = 9$ dBm. DSP included root-raised-cosine (RRC) pulse shaping, necessary equalization, and carrier phase recovery (CPR). To emulate the tracking speed of a practical CPR, a genie-aided (GA) configuration was employed together with a smoothing window. It first extracted instantaneous phase errors by directly comparing the received symbols with the transmitted ones, and then applied a 321-symbol moving average window on these errors before they were used for compensation. Fig. 2(a) depicts the noise analysis results in different configurations with exemplary constellations. In Case 1 or Case 2 with perfect polarization alignment (i.e., the rotation angle $\theta = 0$), while phase noise dominated at zero CD, amplitude noise strength clearly increased as more CD was added. When

CD was sufficiently large, noise in the two dimensions became similarly strong, corresponding to a “circular-shaped” noise distribution in the constellation. On the other hand, when polarization rotation with a non-zero θ was introduced before the Rx SOAs (which was subsequently compensated in DSP), enhanced amplitude noise was observed even without CD, but what interacted with PN in this case was actually the equalization for polarization de-multiplexing rather than CD compensation. As CD increased, the amplitude and phase noise strength also converged to be similar.

Experiment

Experiment was conducted by connecting a pair of coherent optical transmitter and receiver as Fig. 2(b). At the transmitter, 16-QAM symbols were generated from pseudo-random binary sequences and up-sampled, followed by RRC pulse shaping. Multiple over-sampling rates were tested, consequently realizing different symbol rates. The SOAs were integrated within the transmitter in our setup, so to generate EEPN effects, a controlled amount of CD was added to the transmitted signal digitally which was then compensated in receiver DSP. The transmitter frequency response was further pre-compensated before the digital samples were loaded into digital-to-analog converters (DAC). While the SOAs’ operating current was adjusted to generate different transmitter output powers, a variable optical attenuator (VOA) was employed to maintain a constant received power of -10 dBm. At the receiver, signals were captured via analog-to-digital converters (ADC) and processed offline using standard coherent DSP algorithms, with CPR performed using the GA configuration described earlier. Finally, noise of the specific post-DSP symbols was analyzed in amplitude and phase dimensions separately.

Fig. 3 shows the noise RMS values as a function of added CD. Measurements were taken at two transmitter output powers (approximately -1.3 dBm and 6.3 dBm for two polarizations) and three symbol rates (87.5 Gbaud, 109.4 Gbaud and 140 Gbaud). At the lower output power, the amplitude

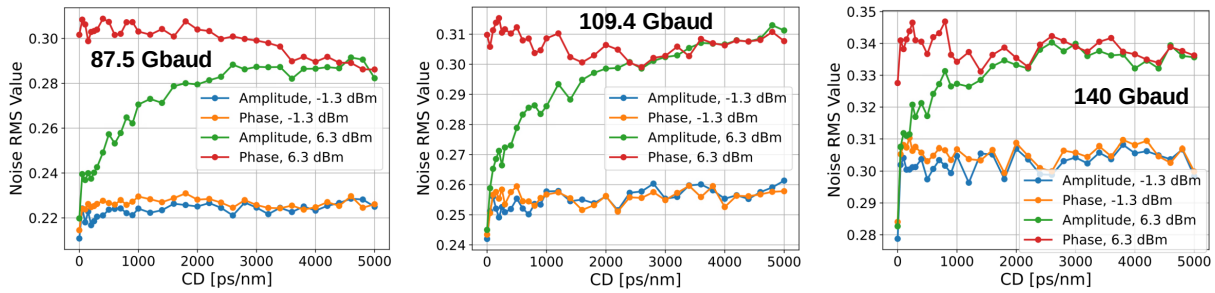


Fig. 3: Measured noise RMS value versus CD with different transmitter output powers and symbol rates.

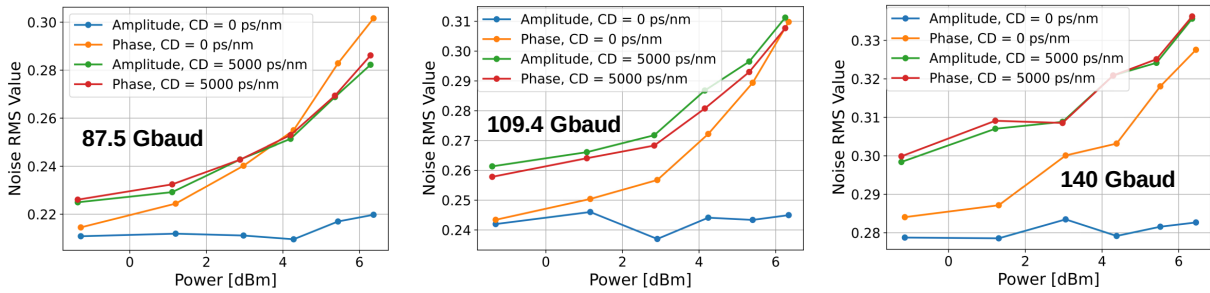


Fig. 4: Measured noise RMS value versus transmitter output power with different added CD and symbol rates.

and phase noise exhibited comparable strength which remained largely insensitive to the added CD, as both SOA nonlinear distortion and its interplay with CD equalization were relatively weak in this case. Note that the slightly higher noise power in the non-zero-CD test cases relative to the zero-CD ones was attributed to the increased signal peak-to-average power ratio (PAPR) and its subsequent impact rather than to EEPN. In contrast, at the higher output power, SOA-induced nonlinear distortion became significant, and the results aligned well with the earlier simulation predictions. Specifically, noise in phase dimension was clearly stronger at zero CD, but as CD increased, more interaction happened between PN and equalization, resulting in a noticeable growth of amplitude noise. Finally, noise in the two dimensions became similarly strong with sufficiently large CD.

Fig. 4 summarizes the noise analysis results with different transmitter output powers and two CD values of zero and 5000 ps/nm. At zero CD, while the amplitude and phase noise strength was comparable in the low-power linear regime, PN became noticeably stronger as the output power increased and more SOA nonlinearity was present. In contrast, with a CD of 5000 ps/nm, EEPN made the amplitude and phase noise have similar strength, and both noise components grew simultaneously when transmitter output power increased.

The experimental results further reveal some differences between SOA-PN-induced and LPN-induced EEPN. First, PN introduced by SOAs generally changes too fast to be fully tracked by normal DSP, and degrades signal quality even before equalization converts it into amplitude noise. This

explains why noticeably more phase errors were still observed after DSP in the zero-CD cases with strong nonlinearity. In contrast, LPN can generally be well compensated by DSP and thus should not affect performance significantly before EEPN is generated. Second, unlike LPN-induced EEPN whose power scales with net CD, the strength of SOA-PN-induced EEPN appears to “saturate” at certain CD values when the dispersion memory length significantly exceeds the correlation length of SOA nonlinearity. Finally, while LPN-induced EEPN intensifies at higher symbol rates, the maximum increment of amplitude noise resulting from SOA-PN-induced EEPN appeared to be larger in our lower-symbol-rate measurements, as shown in Fig. 3. This can be explained by the fact that signal patterns with slower variation tend to cause stronger SOA nonlinearity at the investigated symbol rates [3], potentially leading to more EEPN.

Conclusions

In this work, we provide demonstration of potential EEPN effects arising from SOA nonlinearity. Through combined simulation and experiment analysis, we show that digital equalization, when applied at specific locations within the system, can interact with SOA-induced PN and generate additional amplitude noise. Furthermore, we reveal that such SOA-PN-induced EEPN exhibits a different dependency on key system parameters such as symbol rate and CD when compared with the widely-studied EEPN due to LPN. These findings can offer valuable insights for performance evaluation and impairment mitigation in future coherent systems utilizing SOAs.

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